Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Evaluate a model against measured performance and defend design decisions.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “relative deviation” and “energy budget” to explain your model or result. Sentence starters are optional.

relative deviation: difference from prediction compared with its magnitude

energy budget: an account of all energy stored and transferred

revision: a change made after checking evidence or reasoning

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes 6 times as large.
Compare ideal travel speed and state one real-world limitation.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

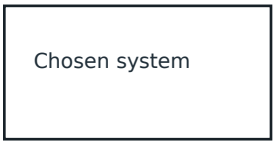
Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw or annotate your final car.
Label three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: 1 2 3

04 Apply the physics

A supplied car covers 5 m in 10 s; its predicted speed was 0.60 m/s. Complete the comparison table.
Find measured speed and use the 0.10 m/s difference to calculate percent deviation relative to the prediction.
State whether it was faster or slower.

Sentence starter: Both ... . They differ in ... because ... .

Distance (m)	Time (s)	Measured v	Predicted v
___	___	___	___

Marzano check - circle: 1 2 3

05 Explain your reasoning

Use your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3

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Continue down each column.

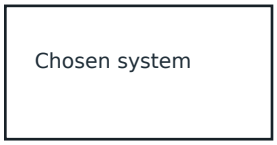
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A device receives 60 J mechanical energy at its shaft and produces electrical energy at its terminals.
Identify it and name one possible loss.
Label input, useful output and other energy transfers.

Sentence starter: The … in my model represents … . Its label shows … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Choose one problem on this sheet.
Circle your current 1, 2 or 3 self-check and name the next step that would improve your explanation.

Sentence starter: My design shows … . It meets the requirement by … .

Problem chosen	What I can explain	Next time
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Propose one redesign and predict two possible effects, including a tradeoff.
State the measurement that would support or challenge your prediction.

Sentence starter: I predict … . My model supports this because … .

Proposed change	Benefit / tradeoff	Test
___	___	___
___	___	___

Marzano check - circle: 1 2 3